

SPECTROSCOPY & FLAME TESTS

Chemistry Series | Comprehensive Study Guide & Worksheet

Name: _____ Class/Sec: _____ Date: _____

1. Key Concepts & Theory Recap

Introduction: Diwali Fireworks & Atomic Light

The brilliant colors in fireworks are chemical clues emitted by metallic elements when heated ! **Spectroscopy** is the study of how light interacts with matter .

The "Jump & Flash Dance" Process:

- Excitation:** Heat energy causes electrons to jump to higher energy levels (excited state) .
- Emission:** Unstable electrons fall back down to their ground state .
- Light Release:** Released energy creates a specific color fingerprint .

Lab Safety & Testing Rules:

- Clean Platinum/Nichrome Wire:** Dip wire in concentrated HCl to prevent contamination.
- Non-Luminous Flame:** Use the hot blue Bunsen burner flame for clear observations.
- Distinct Fingerprints:** Unique electronic arrangements mean unique visual colors !

2. Element & Flame Spectrum Reference Table

Element	Symbol	Flame Colour	Real-World Everyday Sources
Sodium	Na	Bright Yellow	Table salt (NaCl), street lamps, spilled salt on gas burners .
Potassium	K	Lilac / Pale Violet	Fireworks (Sivakasi crackers), plant fertilizers .
Copper	Cu	Blue-Green (Emerald)	Copper water vessels, brass utensils, electrical wires .
Calcium	Ca	Brick Red	Milk, bones, chalk, limestone, festive diyas .
Strontium	Sr	Deep Crimson	Red emergency flares, red fireworks, specialized glass .
Barium *(Added)*	Ba	Apple Green	Green fireworks, medical imaging, drilling fluids.
Lithium *(Added)*	Li	Crimson Red / Magenta	Rechargeable batteries, ceramics, mood stabilizers.

3. Student Practice Questions

Section A: Multiple Choice Questions

Q1. Which element produces a characteristic bright yellow flame when heated?

(a) Copper (Cu)

(b) Sodium (Na)

(c) Calcium (Ca)

(d) Potassium (K)

Q2. What flame colour is observed when potassium salts are introduced into a flame?

(a) Brick Red

(b) Blue-Green

(c) Lilac / Pale Violet

(d) Deep Crimson

Q3. What causes the emission of light during a flame test?

(a) Protons moving between atomic nuclei

(b) Excited electrons returning to lower levels

(c) Creation of new atomic nuclei

(d) Neutrons splitting inside the atom

Q4. *(New)* Why is hydrochloric acid (HCl) used to clean the loop wire before performing a flame test?

(a) To cool the metal down

(b) To convert metals to volatile chlorides and remove impurities

(c) To make the flame burn brighter

(d) To change the element's chemical identity

Section B: Matching & Data Interpretation

Q5. Match each element with its characteristic flame colour:

1. Calcium (Ca) → _____

A. Deep Crimson

3. Sodium (Na) → _____

C. Brick Red

2. Strontium (Sr) → _____

B. Lilac

4. Potassium (K) → _____

D. Bright Yellow

Section C: Short Answer & Conceptual Analysis

Q6. Explain why different metals emit different colors when placed in a burner flame .

Q7. An unknown salt sample from a fireworks factory produces a brilliant blue-green flame . Identify the primary metal ion present and name one everyday object made from this metal .

Q8. *(New)* Compare Continuous Spectrum vs. Line Emission Spectrum. Why are atomic emission spectra referred to as "atomic fingerprints"?